

DAVID WIKSTROM

Simulation & XR Engineer | Technical Lead / Project Manager

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SUMMARY

Simulation and XR engineer and technical lead with 5+ years building physics-based training simulators and real-time 3D applications in Unity, Unreal Engine, and Three.js. Built vehicle dynamics for transit bus simulators and integrated them with physical hardware, including custom cab controls, steering and pedal rigs, and a real bus side console. Lead programmer on 50+ XR solutions, including a Mars Society digital twin with environment-wide Martian gravity and a drivable rover. U.S. Air Force veteran (C-17 Electrical & Environmental Systems).

EXPERIENCE

Lead Extended Reality Developer & Project Manager | MXTreality / Drivers of Tomorrow, Seattle, WA | July 2021 – Present

- Lead developer for Drivers of Tomorrow, MXTreality's bus/CDL training platform for King County Metro and Pierce Transit, spanning VR, mobile, web (Three.js), and simulator hardware, built in Unity, Unreal, and Three.js.
- Engineered physics-based vehicle simulation: friction coefficients, suspension, and acceleration and braking distances tuned across multiple bus variants, with center of mass shifting under passenger load.
- Tuned speed-dependent steering resistance (angular drag that builds with speed and eases at low speed for tight turns), which test drivers said fixed lane drift and over-correction and felt natural.
- Built harsh-braking and harsh-maneuver detection for trainee scoring and feedback.
- Integrated simulators with physical hardware (hardware-in-the-loop): custom bus cab floorboards (accelerator, toggles, switches), Moza steering wheel and pedals, and a real bus side console (parking brake, front doors, engine master switch with Off, Day Run, Night Run, and Night Park).
- Pierce Transit reached a 100% pre-trip inspection pass rate following platform rollout.
- Co-developed MarsVR with The Mars Society (Steam, 2022), a digital twin of the Mars Desert Research Station: environment-wide Martian gravity, a drivable rover with suspension, terrain traction, and low-gravity handling, a Martian running simulation, and side-by-side Earth vs. Mars gravity objects.
- Pushed to modernize the company's older projects and chose Adventure Climb VR, which still had strong reviews. Led its rebuild from a 2019 demo into a full game (engine and physics upgrade, five new levels, complete overhaul), shipped on Steam as the "Brave the Elements Expansion Maps" DLC (July 18, 2022), with the original demo kept as a free first level.
- Lead development and integration across 50+ XR applications in transit training, education, medical/rehab, and assistive technology, using OpenXR, SteamVR, Meta SDK, haptics, and accessibility hardware.
- Project management: build client relationships, gather requirements, set timelines and team goals, and run projects start to finish using Agile, including educational materials, installation, physical demos, and ongoing development and partnerships.
- Write operating instructions, support client installs, and run live client demos with stakeholders.

XR Lab Coordinator | Bellevue College XR Lab, Bellevue, WA | January 2020 – Present

- Coordinate NSF-funded XR Lab operations (NWXR EDU Network): equipment lifecycle, software deployment, faculty/staff training, and equitable student access.
- Supervise student employees and interns; mentor Application Development capstone teams on Unity VR builds using Scrum.
- Lead Unity coding workshops, run XR Lab events and GameJams, and develop lesson plans with faculty.
- Led a rapid-response VR classroom deployment during COVID: a full capstone class in VR in under two weeks (first full VR class session at Bellevue College).

Aircraft Electrical & Environmental Systems Specialist (2A656) | U.S. Air Force, JB McGuire-Dix-Lakehurst, NJ | October 2010 – January 2017

- Maintained electrical, bleed air, and environmental systems on C-17 Globemaster III aircraft (305 AMXS, AMC); \$2.6B fleet.
- Inspected, troubleshoot, and repaired aircraft subsystems using C-17 technical orders and test equipment.
- Previously held a Secret security clearance (currently inactive). Honorably discharged; medical retirement. SrA (E-4).

SKILLS

- **Languages:** C# (primary), C++, Python, JavaScript
- **Engines & frameworks:** Unity, Unreal Engine 5, Three.js, OpenXR, SteamVR, Meta SDK
- **Simulation:** vehicle dynamics, rigid-body physics, gravity and environment modeling, harsh-event detection, hardware-in-the-loop integration
- **Hardware:** haptics, Tesla Suit, steering and pedal rigs, custom control panels, assistive input devices
- **Process & tools:** Agile/Scrum, Jira, Trello, Git, technical documentation, cross-platform build and optimization

EDUCATION

- Bachelor of Applied Science, Information Systems & Technology (Application Development, XR focus), Cum Laude | Bellevue College, Bellevue, WA | January 2018 – June 2021
- Community College of the Air Force | 2010 – 2016 | Military technical training; college credit earned, no degree awarded
- Kudan Institute of Japanese Language | Tokyo, Japan | 2007 | Conversational Japanese

CERTIFICATIONS & AWARDS

- Tesla Suit Certified (company-issued; haptic and sensor suit development)
- Mars Society Outstanding Service Award
- USAF Achievement Medal; USAF Good Conduct Medal
- Vice President, Virtual Reality Club, Bellevue College